Mahmoud Alyosify

Academic CV | PhD Applicant in Artificial Intelligence
Efficient Language-Model Inference · Trustworthy & Evaluable AI Systems
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RESEARCH INTERESTS

Efficient inference for large language models — training-free decoding-time methods, computational and energy cost of generation · **Rigorous evaluation of generative systems** — non-inferiority frameworks, length-controlled judging, reproducibility · **Self-supervised and contrastive representation learning** · **Reinforcement learning for resource-constrained decision-making** · **Multi-agent systems and knowledge-graph-grounded reasoning** · **Empirical software engineering in the age of coding agents**.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Research Project (CISC 898):** *Verbosity-Aware Decoding for Token-Efficient Language Model Inference* — Supervisor: **Dr. Rahatara Ferdousi**.
- Graduate coursework: **Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End Machine Learning Pipelines**.

B.Sc. Computer & Information Science — Bioinformatics — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Undergraduate thesis: **Vitalism Solution** — contactless vital-sign estimation from video using remote photoplethysmography (rPPG) and wavelet-based signal filtering. **Grade: A+**.
- **Semi-Finalist, Microsoft Imagine Cup 2023**; **Finalist, ISEIC 2023** (international research competition).

RESEARCH EXPERIENCE & PROJECTS

Verbosity-Aware Decoding for Token-Efficient Language Model Inference — *2026 – Present*
Master's Research Project (CISC 898), Queen's University · Supervisor: Dr. Rahatara Ferdousi · code

- **Research question:** can generated-token count be reduced at inference time without retraining and without measurable loss of semantic content?
- **Method:** a **training-free LogitsProcessor** fusing stop-token, n-gram-repetition and semantic-redundancy signals under an **entropy-driven gate**; adds no KV-cache or attention overhead and requires no architecture change.
- **Design:** **pre-registered evaluation protocol** testing semantic preservation as a **paired non-inferiority hypothesis** (BERTScore, ROUGE, exact-match/F1, length-controlled LLM judge) — the correct statistical framing for a "no worse than" claim, which conventional significance testing cannot supply.
- **Scope:** **three open-weight model families** (Mistral-7B-Instruct, Llama-3.1-8B-Instruct, Qwen2.5-3B-Instruct) across **four benchmarks** (MT-Bench, YapBench, XSum, SQuAD-v2), reporting token reduction, TTFT / inter-token latency and **GPU energy via NVML**.
- **Result (development set): 25.5–34.9%** token reduction on MT-Bench across all three models, with every pre-registered non-inferiority criterion met.
- **Reproducibility:** deterministic runs with fixed seeds and versioned environments on pinned AWS GPU instances.

Analyzing Review Effort in Human vs. Agentic Pull Requests — *2026*
Empirical Software Engineering · Repository Mining · code

- Built a reproducible mining pipeline over large-scale open-source repositories to quantify whether AI-agent-generated pull requests shift code-review effort relative to human-authored contributions, with explicit cohort definitions and inclusion criteria.

Self-Supervised Contrastive Representation Learning — A Controlled Augmentation Study — *2026*
SimCLR · SupCon · PyTorch · code

- Conducted a **42-experiment augmentation sweep** isolating which transformations drive representation quality, extended with a semi-supervised **SupCon** objective; **84.30% top-1 accuracy** with retrieval evaluation via ONNX/FAISS.

Deep Reinforcement Learning for Cloud Resource Provisioning — *2026*
PPO · DQN · Custom Gymnasium Environment · code

- Designed a custom simulation environment and multi-objective **cost–latency reward**; compared PPO and DQN against static and threshold-based baselines representative of deployed practice.

HORUS Sentinel — Multi-Agent Intelligence Synthesis over a Knowledge Graph *2026*
Multi-Agent Systems · Fine-Tuned LLM · Knowledge Graphs · code

- Architected a swarm of specialised autonomous agents building a **living knowledge graph** as shared memory, over which a self-hosted fine-tuned model produces prioritised, source-traceable reports.

Further Research Output

- **Horus-OSINT** — domain adaptation of **Meta-Llama-3-8B** over **159,826** GTD/GDELT records, deployed on AWS.
- **SCRAP** — predictive modelling of satellite collision risk constrained to a **48-hour** operational lead time.
- **AudioShield** — CNN-based synthetic-speech detection over mel-spectrogram representations.
- **Open research artefacts:** GitHub · Hugging Face · Kaggle · Portfolio.

TEACHING EXPERIENCE

Machine Learning Instructor *Aug 2026 – Sep 2026*
National Telecommunication Institute (NTI) Arish & Assiut, Egypt

- Rehired to teach two **90-hour** applied machine learning bootcamps to **50 trainees**: Arish branch (Aug 2026, online, co-instructor) and Assiut branch (Sep 2026, delivering **59 of 90 hours**).
- Taught linear and logistic regression, SVM, KNN, Naive Bayes, PCA and neural networks through Python labs (NumPy, Pandas, Scikit-learn); wrote the final exams and evaluated end-to-end capstone projects delivered as GitHub repositories and Streamlit apps.

Machine Learning Instructor *Jul 2025 – Sep 2025*
National Telecommunication Institute (NTI) Cairo, Egypt

- Designed and delivered a **90-hour** machine learning program for engineering and computer-science students covering supervised learning, PCA and neural networks with hands-on laboratory sessions.

Online Instructor — Mathematics, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Authored Arabic-language curricula in **discrete mathematics, algorithms, probability, statistics** and **graph data mining**, reaching **10,000+ learners** and **700K+ views**.

PROFESSIONAL EXPERIENCE

AI Engineer Intern *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) Maadi, Cairo, Egypt

- Building an agentic RAG assistant (LangChain, LangGraph, FAISS, FastAPI) with measurement-driven design: retrieval benchmarks (top-1 accuracy **75% to 94%**), evaluation of LLM graders on labelled pairs, and LLM-judge reliability checked against deterministic metrics (0.82 agreement).

AI in Cybersecurity Intern *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) Cairo, Egypt

- Applied AI methods across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles**; co-developed an internal data-management application in **C#, .NET, Entity Framework, SQL Server**.

RESEARCH & TECHNICAL SKILLS

Methodology	Pre-Registered Protocols, Paired Non-Inferiority Testing, Hypothesis Testing, Ablation & Sweep Design, Baseline Construction, Reproducible & Deterministic Experimentation, Scientific Writing
Frameworks	PyTorch, Hugging Face Transformers / PEFT / Datasets, TensorFlow, Scikit-learn, Gymnasium, Optuna, SHAP, NumPy, Pandas, PySpark, FAISS, ONNX, LangChain, LangGraph
Infrastructure	AWS GPU Instances, Docker, FastAPI, CI/CD, Git/GitHub, Linux, NVML Energy Profiling
Programming	Python, C++, C, SQL, C#, Go, R, MATLAB, Bash

AWARDS, SCHOLARSHIPS & SERVICE

- **Digilians Presidential Scholarship**, Queen's University — competitive national AI graduate fellowship (2025).
- **Semi-Finalist**, Microsoft Imagine Cup 2023 · **Finalist**, ISEIC 2023 · **1st Place**, Smart Cities Hackathon 2022 · **1st Place**, Science & Technology Content Competition.
- **Ideal Student Award**, Faculty of Computers & Information, Assiut University, 2022–2023.
- **Service:** Secretary, Higher Scientific & Technological Committee, Student Union, Assiut University · Ambassador, Information Technology Institute (ITI), Assiut · Volunteer Team Leader, USAID University Center for Career Development.
- **Languages:** Arabic (Native) · English (Professional Working Proficiency) · French (Basic). **References available on request.**